

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Process Heat & Material Balance
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-014
Document Revision	R02
Document Status	Issued for Review
Originator	Obinna Joseph
Security Classification	Restricted
Issue Date	20-Dec-22

Process Heat & Material Balance

R02	17-12-22	Issued for Review	O. Joseph	M. Ekpunobi	A. Duru	
R01	08-06-22	Issued for Review	B. Akinwale	K. Yusuff	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	08-06-22	Issued for Client's Review
R02	20-12-22	Re-Issued for Client's Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-014

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
API	American Petroleum Institute
BFPCL	Brass Fertilizer and Petrochemical Company Limited
BOD	Basis Of Design
DEP	Design and Engineering Practice
DPR (now NUPRC)	Department for Petroleum Resources
FEED	Front End Engineering Design
GPP	Gas Processing Plant
MMSCFD	Metric Million Standard Cubic Feet Per Day
MTPD	Metric Tonnes Per Day
NUPRC (formerly DPR)	Nigerian Upstream Petroleum Regulatory Commission
OML	Oil Mining License
PIG	Pipeline Inspection Gauge
PMC	Project Management Consultant
RTU	Remote Telemetry Unit
SPDC	Shell Petroleum Development Company

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

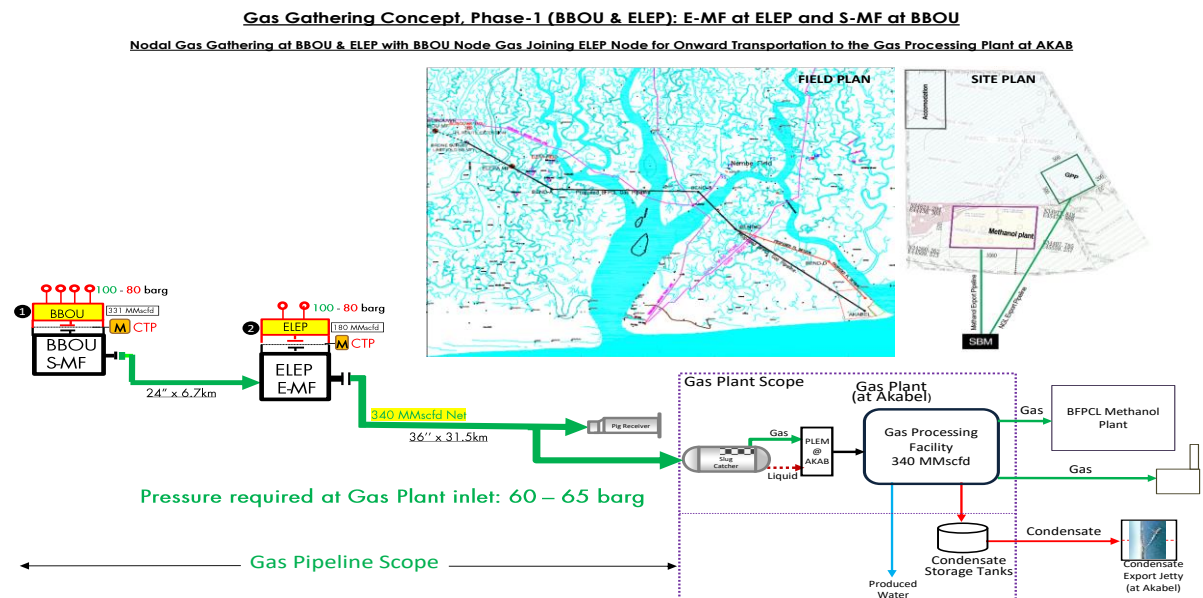


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to present the report of the heat and material balance performed for the FEED for Gas Gathering Pipelines & Associated Infrastructure Phase 1 Project.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended.	This is the principal statute regulating the oil industry. The following regulation relevant to this project is issued pursuant to the Act.
Mineral Oil (Safety) Regulations 1963, as amended.	These regulations, in so far as they apply to pipelines, require compliance with relevant IP, API and ASME codes and standards.
Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)	Provides for licenses to be granted for the establishment and maintenance of pipelines incidental and supplementary to oilfields and oil mining, and for purposes ancillary to such pipelines.
Oil and Gas Pipelines Regulations, 1995	Details the regulations for the design, construction and maintenance of oil and gas pipelines.
Environmental Impact Assessment Act, CAP E12, LFN 2004.	Makes the conduct of an EIA mandatory prior to the development of any project or activity likely to have a significant effect on the environment
Federal Environment Protection Agency Act, CAP F13, LFN 2004.	Provides for the protection of the environment and sustainable development of Nigeria's natural resources.
Environmental Guidelines and Standards For The Petroleum Industry In Nigeria, 1991.	DPR (NUPRC) issued document providing guidelines on pollution abatement and relevant legislation.
EA/SPDC-BFPC UGS-GEP/CSMRPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSMRPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001- 3	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys

40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.
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The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Design Units of Measurement

Variables	Units
Barg	Bar Gauge (Pressure)
Bara	Bar Absolute (Pressure)
bbl	Barrel (Oil Volume)
BCF	Billion Cubic Feet
Bopd	Barrels of Oil per day
Bpd or B/D	Barrel per day
BS&W	Base Sediment and Water
cP	Centipoise (Unit of Viscosity)
F	Fahrenheit
C	Centigrade
km	kilometre
m	meter
Mbopd	Thousand Barrels of Oil per Day
Mbpd	Thousand Barrels of per Day
Mbwpd	Thousand Barrels of Water per Day
MJ	MegaJoule

Variables	Units
mm	Millimetre
MMscfd	Million Standard Cubic Feet Per Day
MW	Mega-Watt
ppm	Part Per Million
psig	Pounds Per Square Inch Gauge (Pressure)
Component	Vol%, mol%, wt ppm
Liquid flow	Kg/h, m ³ /h
Steam flow	Kg/h
Level	%, m, mm
Density	Kg/m ³
Conductivity	μs/cm
Weight	Kg, t
Distance (length)	m, mm
Power	W, KW
Force	N, kN
Current	A, mA
Energy	MJ, kJ
Sound	dBA
Frequency	HZ

3. SIMULATION METHODOLOGY

3.1. Overview

This section describes the process simulation tool, structure and methodology that were used for the determination of the heat and material balances for the project.

This heat and material balance is an output of the process simulation performed to establish the optimum process conditions – material and energy, for efficient delivery of the combined feed gas from BBOU and ELEP to the gas processing plant at Akabel.

The composition of the various fluids to be transported were assumed as provided in the well composition data and their respective source conditions assigned. Simulation was performed using HYSYS software. Process data (obtained from project Basis of Design) were provided into the software for analysis, assuming the correctness of the software's in-built formulas, correlation factors and equation of state among others. Peng Robinson equation of state was selected as the thermodynamic basis for the simulation.

The Peng-Robinson equation of state was used in this simulation because of its high degree of reliability and efficiency in performing Vapor-Liquid Equilibrium (VLE) calculations as well as calculating accurate densities for hydrocarbon systems.

Considering the pressure requirement at delivery point of the Gas Processing Plant, the gas feed inlet pressures were set at approximately 80 bar (g) and 100 bar (g) for both pipelines.

3.2. Well Fluid Simulation

The composition of the well fluids were simulated using the following conditions:

- i. Dry gas composition at each Well
- ii. Reservoir pressure and temperature conditions at each Well
- iii. Estimated flow at from each Well

3.2.1. Dry gas composition at each Well

The gas composition for each field was culled from the Basis of Design Document, EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002. Each of the well fluid composition was assumed to be given on a "dry basis", therefore they were saturated with water before free water and condensate are added in order to arrive at a mixed fluid flow that was used in the simulation of the manifolds and pipelines.

The flashed gas and liquid composition are presented below:

Component	BBOU		ELEP		NUNR		SEIB		OPUG	
	Gas	Liq.	Gas	Liq.	Gas	Liq.	Gas	Liq.	Gas	Liq.
N2	0.07	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.12	0.00
CO2	0.59	0.00	2.30	0.00	0.00	0.00	1.63	0.00	2.53	0.00
H2S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1	93.26	0.00	84.63	0.00	0.00	0.00	83.24	0.00	77.45	0.00
C2	3.17	0.01	6.35	0.03	0.00	0.00	6.27	0.28	9.06	0.13
C3	1.01	0.03	2.67	0.27	0.00	0.00	3.49	1.88	5.40	0.53
iC4	0.61	0.07	0.76	0.37	0.00	0.00	1.20	1.55	2.13	0.76
nC4	0.44	0.07	0.82	0.54	0.00	0.00	1.31	2.36	1.00	1.59
iC5	0.25	0.16	0.43	0.87	0.00	0.00	0.73	2.34	1.07	1.79
nC5	0.19	0.16	0.32	0.68	0.00	0.00	0.50	1.96	0.53	1.97
C6	0.22	1.88	0.59	2.26	0.00	0.00	0.68	4.76	0.53	5.23
C7	0.17	6.21	0.69	6.75	0.00	0.00	0.62	9.71	0.17	8.16
C8	0.02	6.61	0.44	12.76	0.00	0.00	0.28	15.65	0.05	16.75
C9	0.00	7.13	0.00	10.87	0.00	0.00	0.00	11.71	0.00	8.80
C10	0.00	5.70	0.00	8.62	0.00	0.00	0.00	9.14	0.00	5.87
C11	0.00	7.83	0.00	6.64	0.00	0.00	0.00	5.68	0.00	6.44

					0	0				
C12+	0.00	64.14	0.00	49.34	0.0	0.0	0.00	33.00	0.00	42.00
					0	0				
Total	100.0	100.0	100.0	100.0	0.0	0.0	100.0	100.0	100.0	100.0
	0	0	0	0	0	0	0	0	0	0

The Peng Robinson EOS was selected for the VLE calculations because it is the generally recommended model For Oil, Gas, or Petrochemical applications. The PR property package rigorously solves any single-, two-, or three-phase system with a high degree of efficiency and reliability and is applicable over a wide range of conditions:

- Temperature Range > -271 °C or -456 °F
- Pressure Range < 100,000 kPa or 15,000 psia

The well fluids compositions fall within these range.

3.2.2. Reservoir pressure and temperature conditions at each Well

The table below shows the reservoir pressure and temperature used for each Well.

	BBOU	ELEP	NUNR	SEIB	OPUG
Reservoir Pressure, psia	3,246	6,910	7,259	6,533	5,176
Reservoir Temp, deg. C	78	123	102	117	117

3.2.3. Estimated flow at from each Well

The two Pipelines (BBOU-to-Elepa and Elepa-to-Akabel) were simulated using the maximum combined flow of gas, condensate and water that will be handled by each of the Pipeline – BBOU and Elepa.

BBOU Manifold. This is a very conservative assumptions made since the actual forecast are not available at the time of preparing this report.

3.2.3.1. Selected Flow for BBOU-to-Elepa Pipeline

The maximum flow used for gas, condensate and water for the BBOU-to-Elepa Pipeline occurred in Years 9, 18 and 7 respectively as highlighted in the table below. Maximum combined flow are 331 MMSCFD gas, 16226 bpd of condensate and 9002 bpd of water.

Year	TOTAL FLUID RATE			BUBOUWE BOU			OPUGBENE		
	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
1	181	944	907	181	944	907	0	0	0
2	181	878	907	181	878	907	0	0	0
3	181	788	907	181	788	907	0	0	0
4	181	742	7,855	181	742	7,855	0	0	0
5	181	707	7,547	181	707	7,547	0	0	0
6	215	840	8,491	215	840	8,491	0	0	0
7	305	1,217	9,002	305	1,217	9,002	0	0	0
8	327	1,323	8,193	327	1,323	8,193	0	0	0
9	331	1,338	8,038	331	1,338	8,038	0	0	0
10	93	1,763	2,527	67	239	2,397	26	1,525	129
11	75	3,242	1,301	21	59	1,031	54	3,183	270
12	74	3,457	1,074	16	42	784	58	3,416	290
13	71	4,084	1,513	2	4	85	69	4,080	1,429
14	71	4,161	3,048	0	0	0	71	4,161	3,048
15	94	5,541	3,748	0	0	0	94	5,541	3,748
16	154	9,082	3,859	0	0	0	154	9,082	3,859
17	170	12,115	3,148	0	0	0	170	12,115	3,148
18	174	16,226	1,667	0	0	0	174	16,226	1,667
19	177	15,812	1,558	0	0	0	177	15,812	1,558
20	182	15,606	2,775	0	0	0	182	15,606	2,775

3.2.3.2. Selected Flow for Elepa-to-Akabel Pipeline

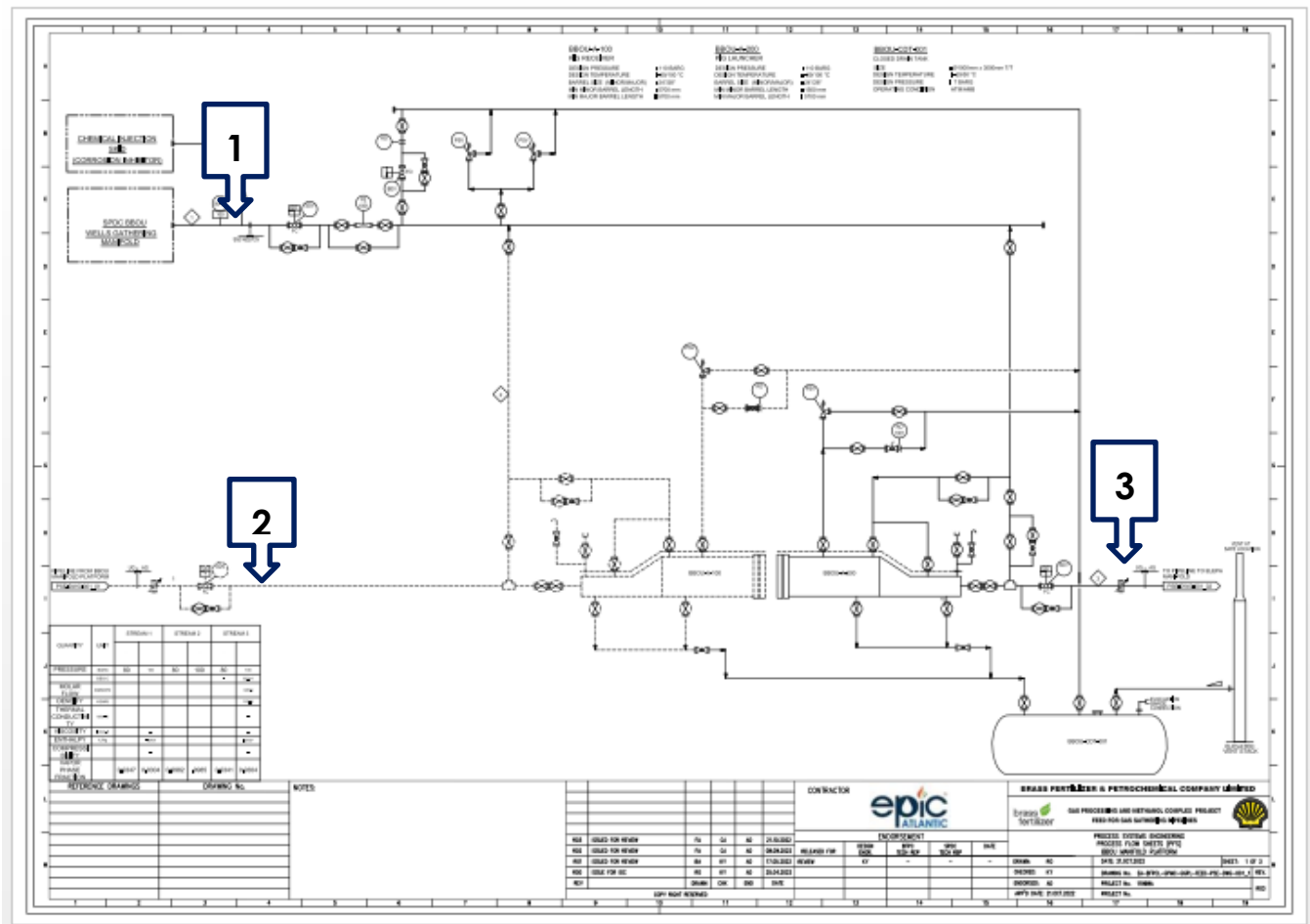
year	TOTAL FLUID RATE			BUBOUWE BOU			ELEPA			NUN RIVER			SEIB			OPUG		
	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
1	361	11,032	1,807	181	944	907	180	10,088	900	0	0	0	0	0	0	0	0	0
2	361	10,965	1,807	181	878	907	180	10,088	900	0	0	0	0	0	0	0	0	0
3	361	10,876	1,807	181	788	907	180	10,088	900	0	0	0	0	0	0	0	0	0
4	361	10,830	15,835	181	742	7,855	180	10,088	7,980	0	0	0	0	0	0	0	0	0
5	361	10,794	15,213	181	707	7,547	180	10,088	7,666	0	0	0	0	0	0	0	0	0
6	360	8,985	13,393	215	840	8,491	145	8,146	4,902	0	0	0	0	0	0	0	0	0
7	357	4,128	11,599	305	1,217	9,002	52	2,911	2,597	0	0	0	0	0	0	0	0	0
8	356	2,953	9,647	327	1,323	8,193	29	1,630	1,454	0	0	0	0	0	0	0	0	0
9	356	2,724	9,275	331	1,338	8,038	25	1,386	1,236	0	0	0	0	0	0	0	0	0
10	382	18,828	4,837	67	239	2,397	19	1,076	960	244	14,464	1,221	26	1,525	129	26	1,525	129
11	385	21,637	3,357	21	59	1,031	11	628	561	245	14,583	1,225	54	3,183	270	54	3,183	270
12	385	21,933	3,015	16	42	784	9	478	426	245	14,583	1,225	58	3,416	290	58	3,416	290
13	386	22,799	13,832	2	4	85	1	52	46	245	14,583	10,844	69	4,080	1,429	69	4,080	1,429
14	386	22,905	16,513	0	0	0	0	0	0	245	14,583	10,417	71	4,161	3,048	71	4,161	3,048
15	384	21,996	14,149	0	0	0	0	0	0	196	10,914	6,653	94	5,541	3,748	94	5,541	3,748
16	379	22,323	11,246	0	0	0	0	0	0	71	4,159	3,528	154	9,082	3,859	154	9,082	3,859
17	380	26,558	8,272	0	0	0	0	0	0	40	2,329	1,976	170	12,115	3,148	170	12,115	3,148
18	382	34,433	5,014	0	0	0	0	0	0	34	1,980	1,680	174	16,226	1,667	174	16,226	1,667
19	380	33,161	4,421	0	0	0	0	0	0	26	1,538	1,304	177	15,812	1,558	177	15,812	1,558
20	379	32,111	6,313	0	0	0	0	0	0	15	898	762	182	15,606	2,775	182	15,606	2,775

4. Results

The results of the heat and material balance calculations performed using the ASPEN HYSYS Software for selected positions in the process line are captured in this section. The points of interest are shown in the Process Flow Scheme Drawing, document EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-001.

4.1. Heat and Material Balance for BBOU Manifold

The locations where heat and material balances were taken are shown in the figure below as streams 1, 2 and 3. The results of each of the stream are shown in following paragraphs for both the low (80 barg supply pressure) and high pressure (100 barg supply pressure) conditions.

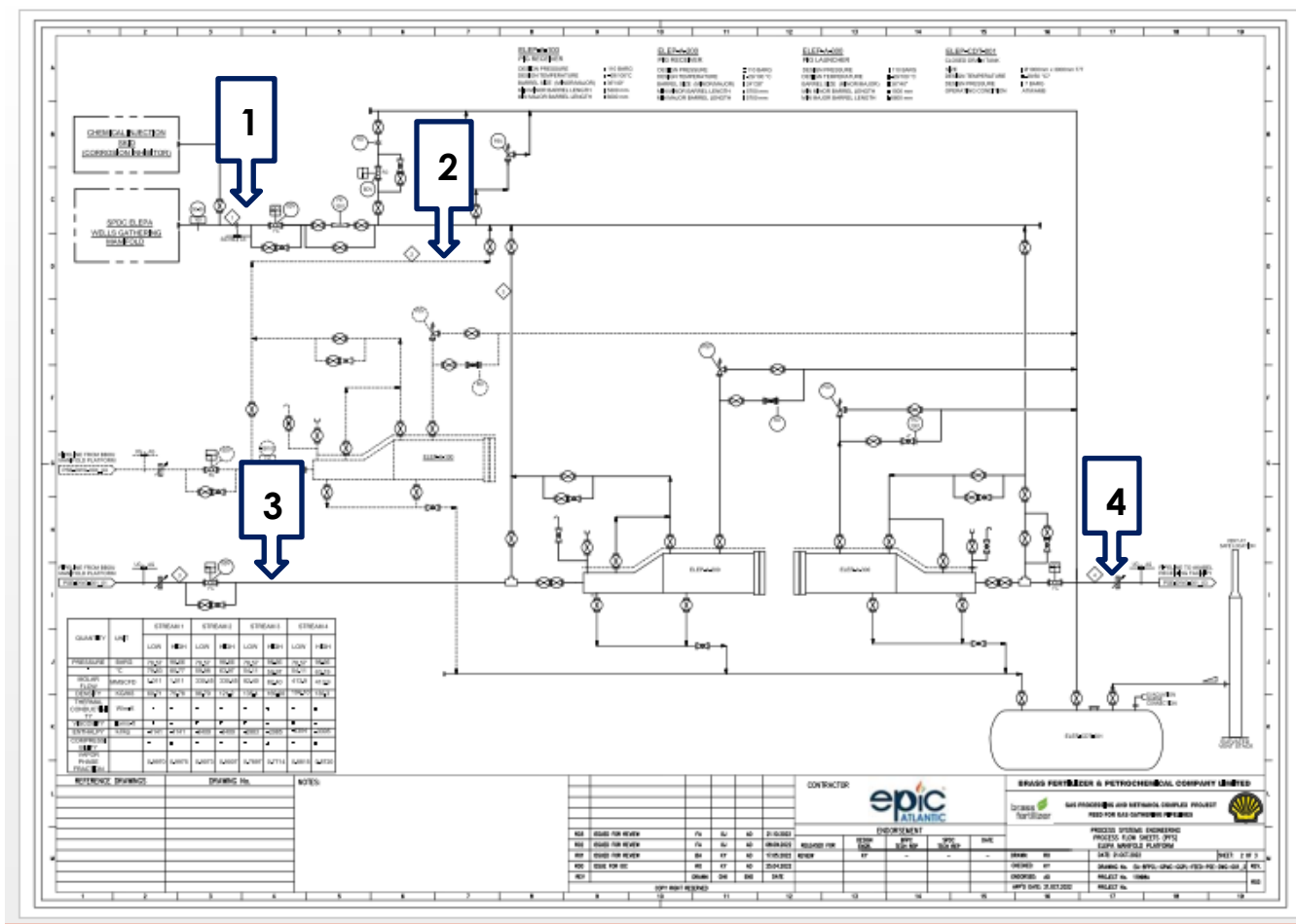


BBOU MANIFOLD (331 MMSCFD)						
PARAMETER	STREAM 1		STREAM 2		STREAM 3	
Pressure Delivery	LOW	HIGH	LOW	HIGH	LOW	HIGH
Vapour Fraction	0.9791	0.9783	0.0000	0.0000	0.9346	0.9303
Temperature [C]	52.16	57.59	85.47	84.47	57.94	62.14
Pressure [bar]	80.00	100.00	80.00	100.00	80.00	100.00
Molar Flow [MMSCFD]	336	336	10	10	346	346
Mass Flow [kg/h]	342339	342339	76981	76981	419320	419320
Liquid Volume Flow [barrel/day]	148253	148253	15681	15681	163935	163935
Heat Flow [kJ/h]	-1405321197	-1405321197	-149011120	-149011120	-1554332317	-1554332317
COMPOSITION (MOLE BASIS)						
Nitrogen	0.0007	0.0007	0.0000	0.0000	0.0007	0.0007
CO2	0.0058	0.0058	0.0000	0.0000	0.0057	0.0057
Methane	0.9199	0.9199	0.0000	0.0000	0.8933	0.8933
Ethane	0.0313	0.0313	0.0001	0.0001	0.0304	0.0304
Propane	0.0100	0.0100	0.0003	0.0003	0.0097	0.0097
i-Butane	0.0060	0.0060	0.0007	0.0007	0.0059	0.0059
n-Butane	0.0043	0.0043	0.0007	0.0007	0.0042	0.0042
i-Pentane	0.0025	0.0025	0.0016	0.0016	0.0024	0.0024
n-Pentane	0.0019	0.0019	0.0016	0.0016	0.0019	0.0019
n-Hexane	0.0022	0.0022	0.0188	0.0188	0.0027	0.0027
n-Heptane	0.0017	0.0017	0.0621	0.0621	0.0034	0.0034
n-Octane	0.0002	0.0002	0.0661	0.0661	0.0021	0.0021
n-Nonane	0.0000	0.0000	0.0713	0.0713	0.0021	0.0021
n-Decane	0.0000	0.0000	0.0570	0.0570	0.0017	0.0017

n-C11	0.0000	0.0000	0.0783	0.0783	0.0023	0.0023
n-C12	0.0000	0.0000	0.6414	0.6414	0.0186	0.0186
H2O	0.0035	0.0035	0.0000	0.0000	0.0034	0.0034
n-C13	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
n-C14	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
n-C15	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
n-C16	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
n-C17	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
n-C18	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
n-C19	0.0101	0.0101	0.0000	0.0000	0.0098	0.0098
n-C20	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

4.2. Heat and Material Balance for Elepa Manifold

The locations where heat and material balances were taken are shown in the figure below as streams 1, 2, 3 and 4. The results of each of the stream are shown in following paragraphs for both the low (80 barg supply pressure) and high pressure (100 barg supply pressure) conditions.

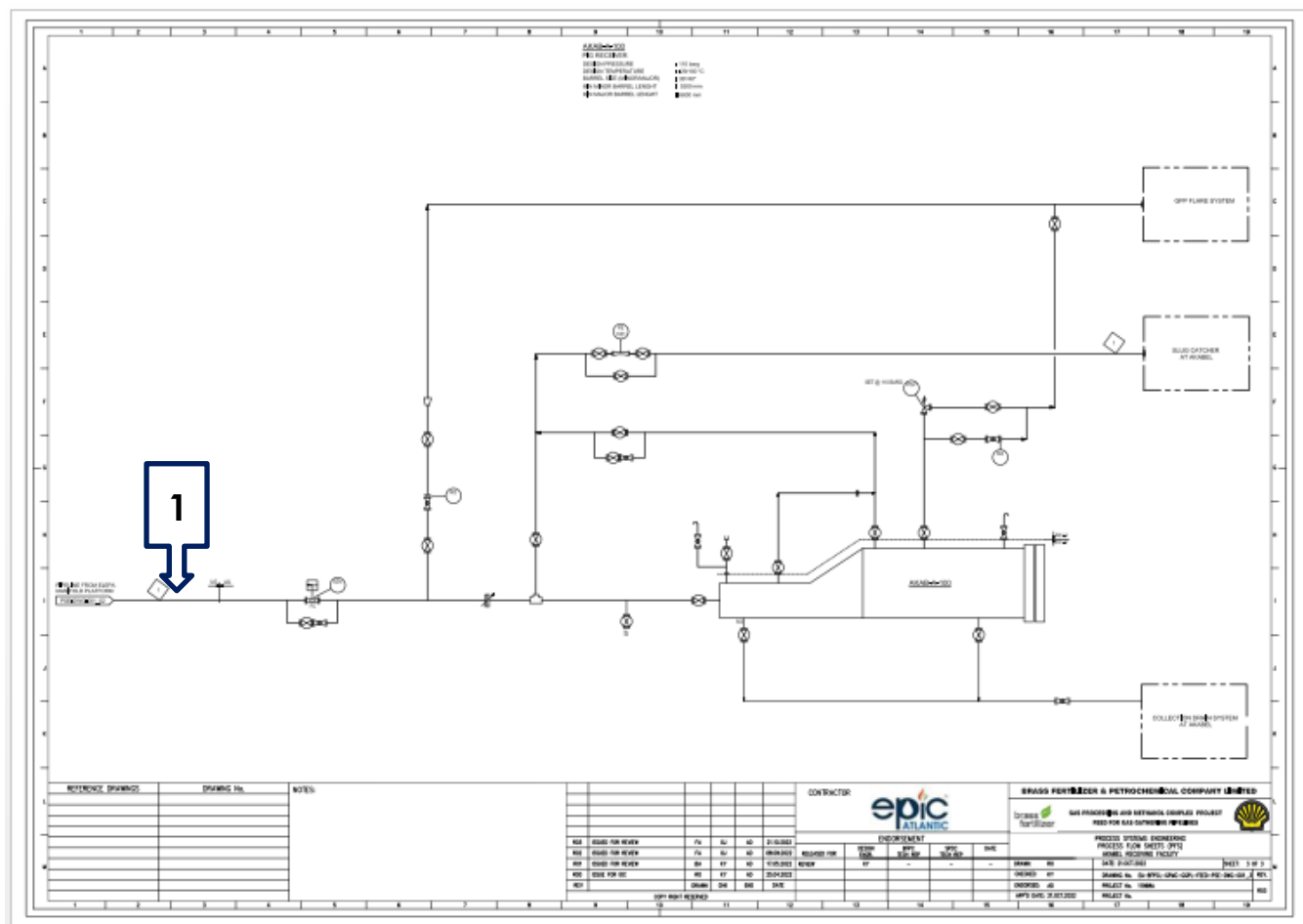


ELEPA MANIFOLD (386 MMSCFD)								
PARAMETER	STREAM 1		STREAM 2		STREAM 3		STREAM 4	
Pressure Delivery	LOW	HIGH	LOW	HIGH	LOW	HIGH	LOW	HIGH
Vapour Fraction	0.9971	0.9975	0.9073	0.9007	0.7897	0.7714	0.8815	0.8720
Temperature [C]	80.28	86.14	59.98	63.97	57.84	59.57	59.75	63.19
Pressure [bar]	80.00	100.00	78.57	98.66	78.57	98.66	78.57	98.66
Molar Flow [MMSCFD]	1	1	330	330	82	82	414	414
Mass Flow [kg/h]	1004	1004	468647	468647	156043	156043	625695	625695
Liquid Volume Flow [barrel/day]	439	439	168283	168283	48561	48561	217284	217284
Heat Flow [kJ/h]	-4.16E+06	-4.16E+06	-1.60E+09	-1.60E+09	-4.65E+08	-4.66E+08	-2.07E+09	-2.07E+09
COMPOSITION (MOLE BASIS)								
Nitrogen	0.0107	0.0107	0.0103	0.0103	0.0006	0.0006	0.0083	0.0083
CO ₂	0.0114	0.0114	0.0109	0.0109	0.0051	0.0051	0.0098	0.0098
Methane	0.8505	0.8505	0.8167	0.8167	0.8035	0.8035	0.8142	0.8142
Ethane	0.0623	0.0623	0.0599	0.0599	0.0273	0.0273	0.0534	0.0534
Propane	0.0234	0.0234	0.0226	0.0226	0.0087	0.0087	0.0198	0.0198
i-Butane	0.0063	0.0063	0.0062	0.0062	0.0053	0.0053	0.0060	0.0060
n-Butane	0.0065	0.0065	0.0064	0.0064	0.0039	0.0039	0.0059	0.0059
i-Pentane	0.0042	0.0042	0.0042	0.0042	0.0023	0.0023	0.0038	0.0038
n-Pentane	0.0016	0.0016	0.0017	0.0017	0.0018	0.0018	0.0017	0.0017
n-Hexane	0.0043	0.0043	0.0047	0.0047	0.0042	0.0042	0.0046	0.0046
n-Heptane	0.0050	0.0050	0.0068	0.0068	0.0090	0.0090	0.0072	0.0072
n-Octane	0.0033	0.0033	0.0072	0.0072	0.0082	0.0082	0.0074	0.0074
n-Nonane	0.0000	0.0000	0.0037	0.0037	0.0087	0.0087	0.0047	0.0047
n-Decane	0.0000	0.0000	0.0029	0.0029	0.0069	0.0069	0.0037	0.0037

n-C11	0.0000	0.0000	0.0022	0.0022	0.0095	0.0095	0.0036	0.0036
n-C12	0.0000	0.0000	0.0020	0.0020	0.0779	0.0779	0.0171	0.0171
H2O	0.0105	0.0105	0.0035	0.0035	0.0030	0.0030	0.0034	0.0034
n-C13	0.0000	0.0000	0.0020	0.0020	0.0000	0.0000	0.0016	0.0016
n-C14	0.0000	0.0000	0.0019	0.0019	0.0000	0.0000	0.0015	0.0015
n-C15	0.0000	0.0000	0.0016	0.0016	0.0000	0.0000	0.0013	0.0013
n-C16	0.0000	0.0000	0.0013	0.0013	0.0000	0.0000	0.0010	0.0010
n-C17	0.0000	0.0000	0.0010	0.0010	0.0000	0.0000	0.0008	0.0008
n-C18	0.0000	0.0000	0.0009	0.0009	0.0000	0.0000	0.0007	0.0007
n-C19	0.0000	0.0000	0.0160	0.0160	0.0139	0.0139	0.0155	0.0155
n-C20	0.0000	0.0000	0.0034	0.0034	0.0000	0.0000	0.0027	0.0027

4.3. Heat and Material Balance for Akabel Receiving Facility

The locations where heat and material balances were taken are shown in the figure below as streams 1, 2, 3 and 4. The results of each of the stream are shown in following paragraphs for both the low (80 barg supply pressure) and high pressure (100 barg supply pressure) conditions.

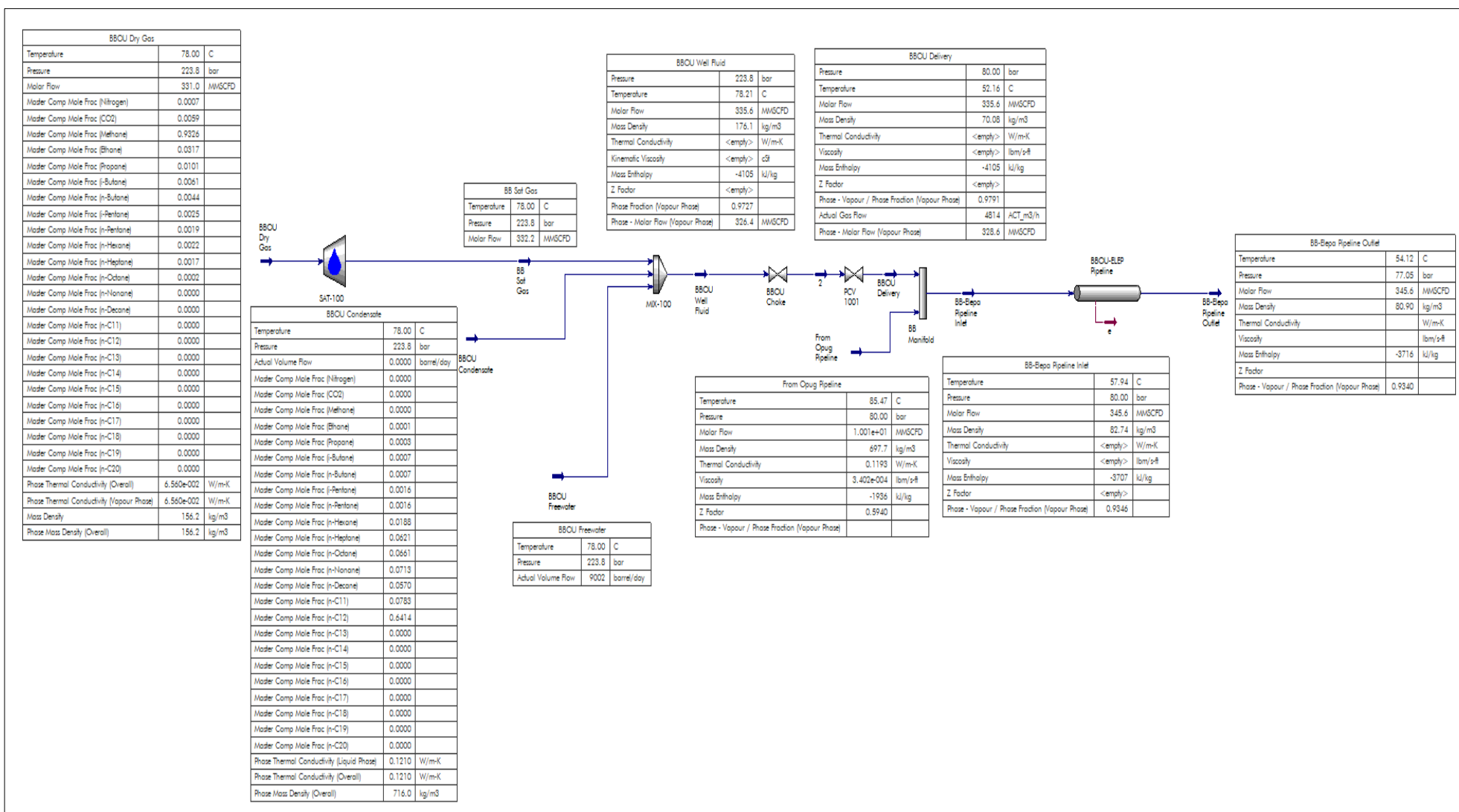


AKABEL RECEIVING FACILITY		
PARAMETER	STREAM 1	
Pressure Delivery	LOW	HIGH
Vapour Fraction	0.8753	0.8620
Temperature [C]	37.65	39.26
Pressure [bar]	66.29	87.59
Molar Flow [MMSCFD]	414	414
Mass Flow [kg/h]	625695	625695
Liquid Volume Flow [barrel/day]	217284	217284
Heat Flow [kJ/h]	-2100461095	-2105637472

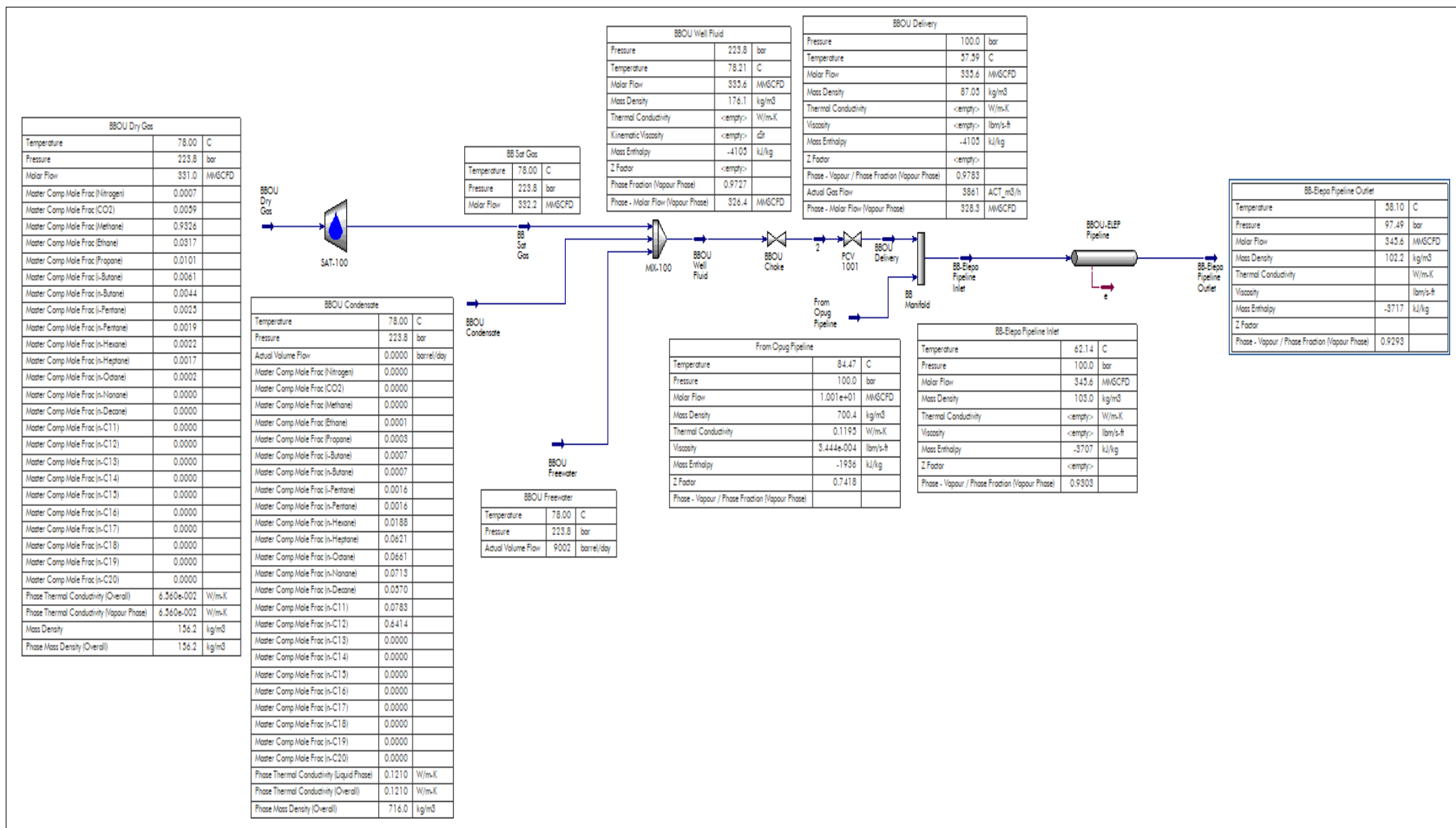
COMPOSITION (MOLE BASIS)		
Nitrogen	0.0083	0.0083
CO ₂	0.0098	0.0098
Methane	0.8142	0.8142
Ethane	0.0534	0.0534
Propane	0.0198	0.0198
i-Butane	0.0060	0.0060
n-Butane	0.0059	0.0059
i-Pentane	0.0038	0.0038
n-Pentane	0.0017	0.0017
n-Hexane	0.0046	0.0046
n-Heptane	0.0072	0.0072
n-Octane	0.0074	0.0074
n-Nonane	0.0047	0.0047
n-Decane	0.0037	0.0037
n-C11	0.0036	0.0036
n-C12	0.0171	0.0171
H ₂ O	0.0034	0.0034
n-C13	0.0016	0.0016
n-C14	0.0015	0.0015
n-C15	0.0013	0.0013
n-C16	0.0010	0.0010
n-C17	0.0008	0.0008
n-C18	0.0007	0.0007
n-C19	0.0155	0.0155
n-C20	0.0027	0.0027

5. Appendix

5.1. HYSYS Simulation PFD Snapshot for BBOU Low Pressure Case



5.2. HYSYS Simulation PFD Snapshot for BBOU High Pressure Case



5.4. HYSYS Simulation PFD Snapshot for Elepa High Pressure Case

